Exercise protein alignment

1. Ortholog VS paralog
   - Compare alignment result of human HBA1, HBB to that of human HBB and mouse HBB
2. MSA
   - Take human HBA
   - Find orthologs from mouse, dog, chicken, zebrafish, salmon (and more) with eggNOG (think about which taxonomic level they share)
   - Align them with Clustal Omega

Human HBA1, HBB

string-db.org

Sequence alignment:

ENSP00000494175	**HBB**	Homo sapiens	Hemoglobin subunit beta; Invol...
ENSP00000322421	**HBA1**	Homo sapiens	Hemoglobin subunit alpha 1.

alignment score 114.8 bits.

```
ENSP0000049 4        LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLS
                     L+P +K+ V A WGKV  +  E G EAL R+ + +P T+ +F  F DLS
ENSP0000032 3        LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLS


ENSP0000049 51       TPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHV
                           G+ +VK HGKKV  A ++ +AH+D++    + LS+LH  KL V
ENSP0000032 51       H-----GSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRV


ENSP0000049 100      DPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH
                     DP NF+LL + L+  LA H   EFTP V A+  K +A V+  L  KY
ENSP0000032 95       DPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR


//
```

Please note: In order not to miss anything, STRING has a very low score-cutoff for reporting protein similarities. This means that a significant number of the reported hits are expected to be spurious. This is not a problem for detecting co-occurrence links (because of our singular value decomposition approach and the benchmarking). However, when looking for homologs of a specific protein in a particular species, make sure you have inspected the above alignment and found it relevant. As a rule-of-thumb, bitscores below 60 may indicate spurious hits.

Human HBB, mouse HBB

string-db.org

Sequence alignment:

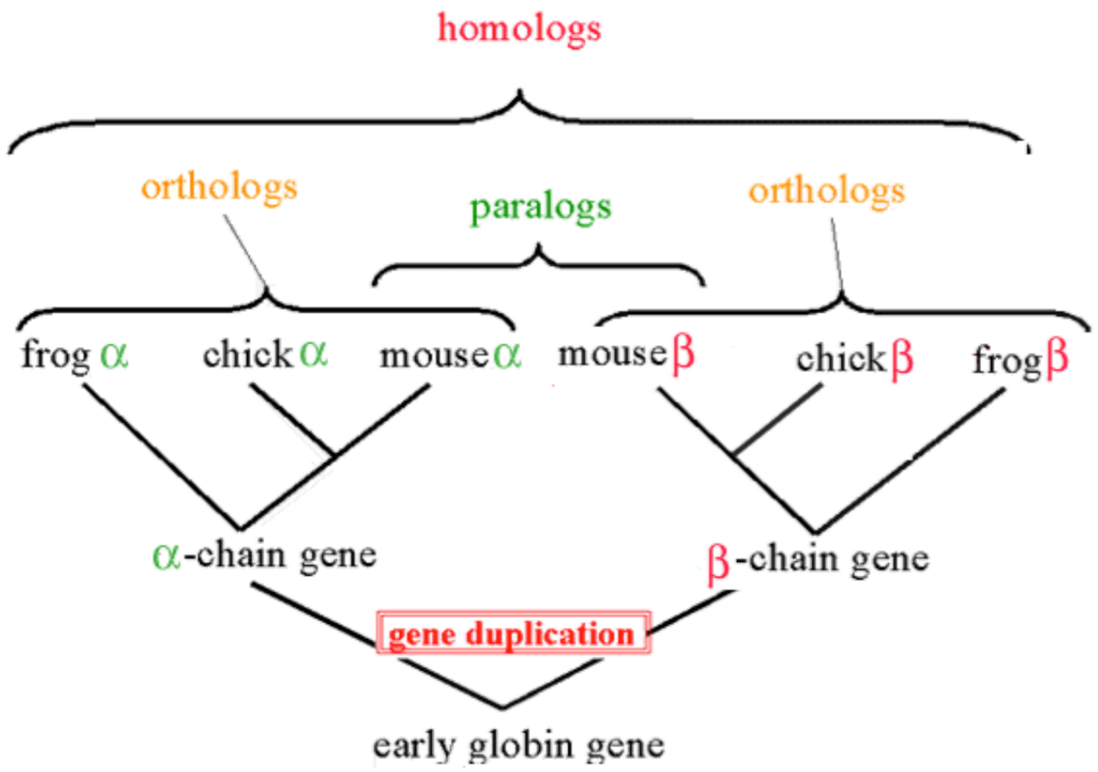
ENSP00000494175	**HBB**	Homo sapiens	Hemoglobin subunit beta; Inv...
ENSMUSP00000095794	**Hbb-bt**	Mus musculus	Hemoglobin, beta adult t chai...

alignment score 250.8 bits.

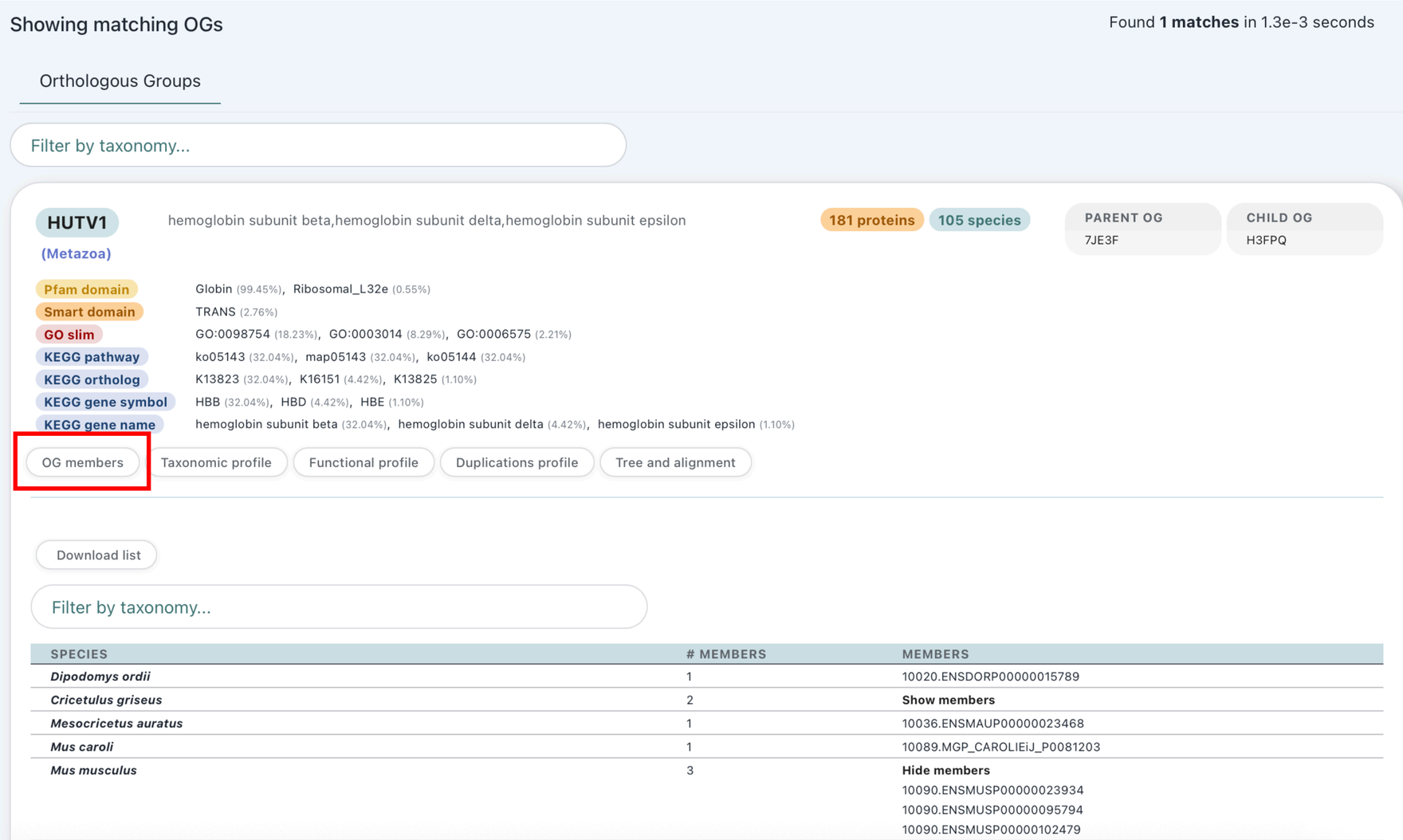
```
ENSP0000049 1        MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDL
                     MVHLT  EK+AV+ LWGKVN DEVGGEALGRLLVVYPWTQR+F+SFGDL
ENSMUSP0000 1        MVHLTDAEKAAVSGLWGKVNADEVGGEALGRLLVVYPWTQRYFDSFGDL


ENSP0000049 50       STPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLH
                     S+  A+MGN KVKAHGKKV+ AF+DGL HLD+LKGTFA+LSELHCDKLH
ENSMUSP0000 50       SSASAIMGNAKVKAHGKKVITAFNDGLNHLDSLKGTFASLSELHCDKLH


ENSP0000049 99       VDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH
                     VDPENFRLLGN++V VL HH GK+FTP  QAA+QKVVAGVA ALAHKYH
ENSMUSP0000 99       VDPENFRLLGNMIVIVLGHHLGKDFTPAAQAAFQKVVAGVAAALAHKYH


ENSP0000049 148

ENSMUSP0000 148


//
```

Please note: In order not to miss anything, STRING has a very low score-cutoff for reporting protein similarities. This means that a significant number of the reported hits are expected to be spurious. This is not a problem for detecting co-occurrence links (because of our singular value decomposition approach and the benchmarking). However, when looking for homologs of a specific protein in a particular species, make sure you have inspected the above alignment and found it relevant. As a rule-of-thumb, bitscores below 60 may indicate spurious hits.

Indeed, alpha-beta separation happen before human-mouse speciation



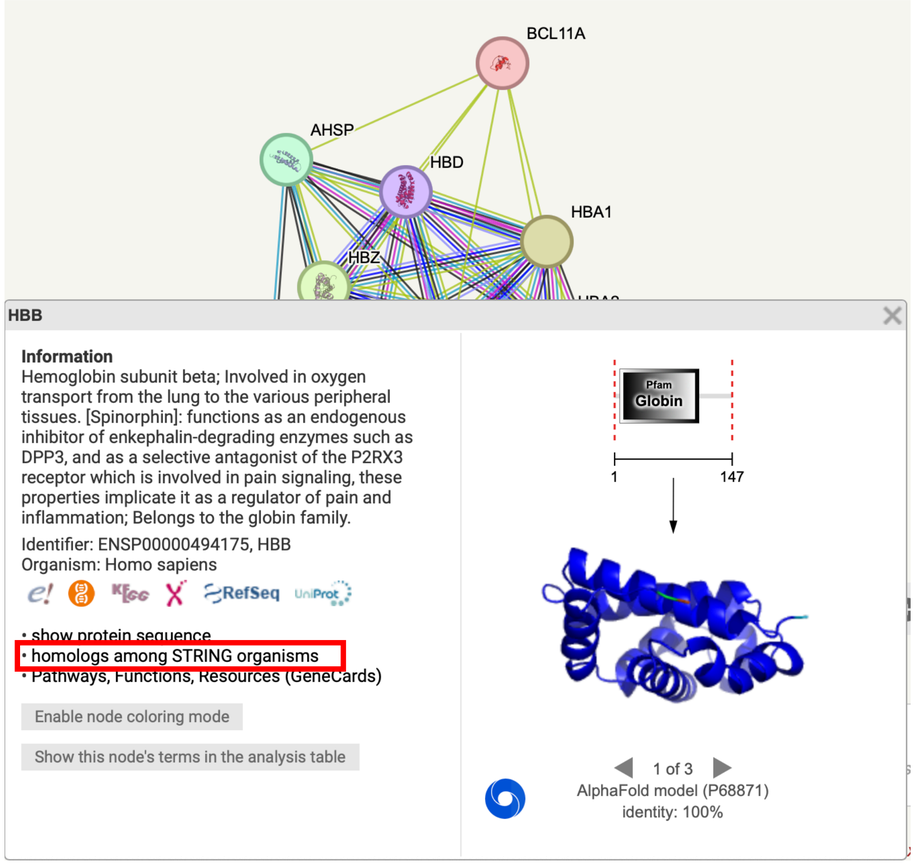
- Search protein
- Choose taxonomic level
- Find orthologous protein sequence
- NOTE: choose the closest shared taxonomic level. e.g. Primates (not eukaryota) for human&monkey



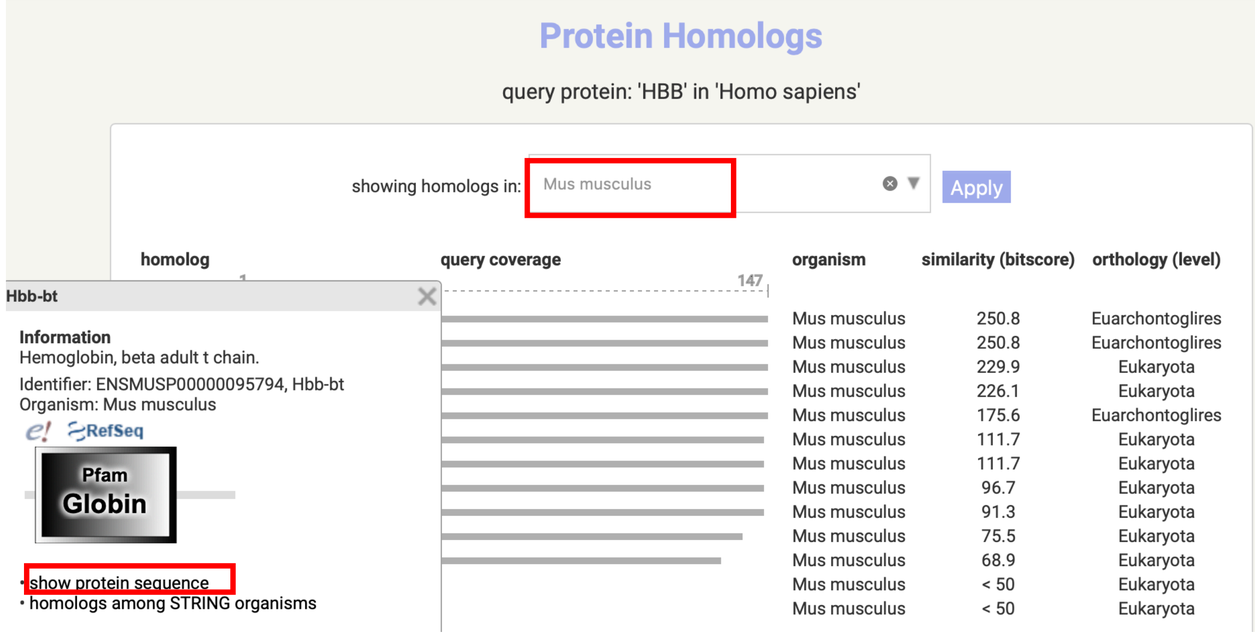
STRING known and predicted protein-protein interactions

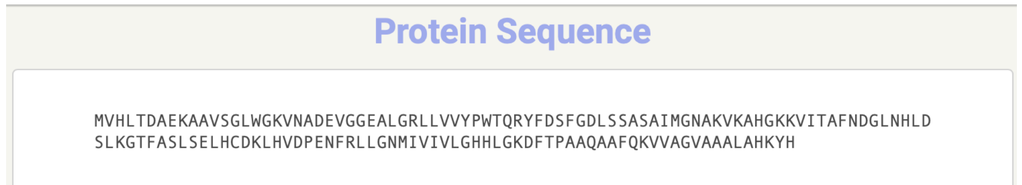
currently covers 59'309'604 proteins from 12'535 organisms.

- Search protein → click protein ball



- Search species → click protein name





Clustal OMEGA

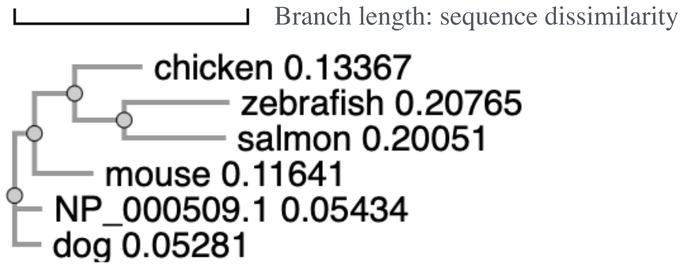
```
chicken         ------------------------------------------------------------   0
mouse           ------------------------------------------------------------   0
NP_000509.1     ------------------------------------------------------------   0
dog             MVHFSAEEKAAITSLWGKVNVEDAGGEALGRLLVVYPWTQRFFDNFGNLSSESAIMGNPK   60
zebrafish       ------------------------------------------------------------   0
salmon          ------------------------------------------------------------   0


chicken         ---------------------------------------------------MVHWTAEEK   9
mouse           ---------------------------------------------------MVHLTDAEK   9
NP_000509.1     ---------------------------------------------------MVHLTPEEK   9
dog             VKAHGKKVLTSFGNAIKHMDDLKDTFAELSELHCDKLHVDPENFKQPQTDTMVHLTAEEK   120
zebrafish       ---------------------------------------------------MVEWTDAER   9
salmon          ---------------------------------------------------MVDWTDAEK   9
                                                                    **. *  *:

chicken         QLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSSPTAILGNPMVRAHGKKVL   69
mouse           AAVSGLWGKVNADEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAKVKAHGKKVI   69
NP_000509.1     SAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVL   69
dog             SLVSGLWGKVNVDEVGGEALGRLLIVYPWTQRFFDSFGDLSTPDAVMSNAKVKAHGKKVL   180
zebrafish       TAILGLWGKLNIDEIGPQALSRCLIVYPWTQRYFATFGNLSSPAAIMGNPKVAAHGRTVM   69
salmon          STISAVWGKVDINEVGPLALARVLIVYPWTQRYFGSFGDVSTPAAIMGNPKVAAHGKVVC   69
                  : .:***::  * *  **.* *:*******:* :**::*:  *::.*  * ***: *

chicken         TSFGDAVKNLDNIKNTFSQLSELHCDKLHVDPENFRLLGDILIIVLAAHFSK-DFTPECQ   128
mouse           TAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMIVIVLGHHLGK-DFTPAAQ   128
NP_000509.1     GAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGK-EFTPPVQ   128
dog             NSFSDGLKNLDNLKGTFAKLSELHCDKLHVDPENFKLLGNVLVCVLAHHFGK-EFTPQVQ   239
zebrafish       GGLERAIKNMDNVKNTYAALSVMHSEKLHVDPDNFRLLADCITVCAAMKFGQAGFNADVQ   129
salmon          GALDKAVKNMGNILATYKSLSETHANKLFVDPDNFRVLADVLTIVIAAKFGA-SFTPEIQ   128
                 .:  .: ::..:  *:  **  *.:**.***:**::*.: :    . ::.   *.   *

chicken         AAWQKLVRVVAHALARKYH    147
mouse           AAFQKVVAGVAAALAHKYH    147
NP_000509.1     AAYQKVVAGVAN-------    140
dog             AAYQKVVAGVANALAHKYH    258
zebrafish       EAWQKFLAVVVSALCRQYH    148
salmon          ATWQKFMKVVVAAMGSRYF    147
                 ::**.:  *.
```

* fully conserved (same amino acid)

: strongly similar (conservative substitution)

. weakly similar

no similarity



To upload to course-results

- Table of ClinVar
- UniProt exercise (one report per group)
- Superposition of HBB wild-type and HBB mutant from sickle cell anemia